

Machine Learning

CSE343/CSE543/ECE363/ECE563

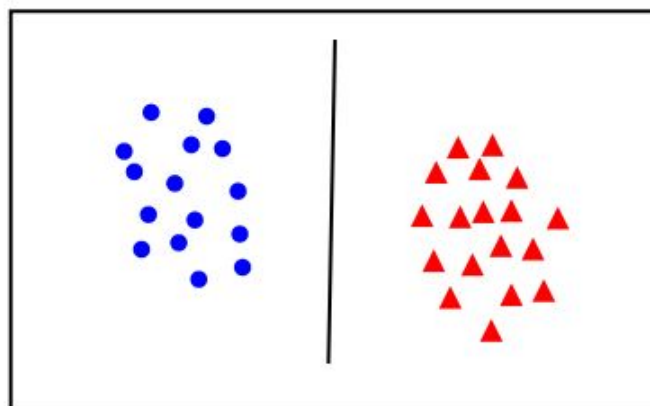
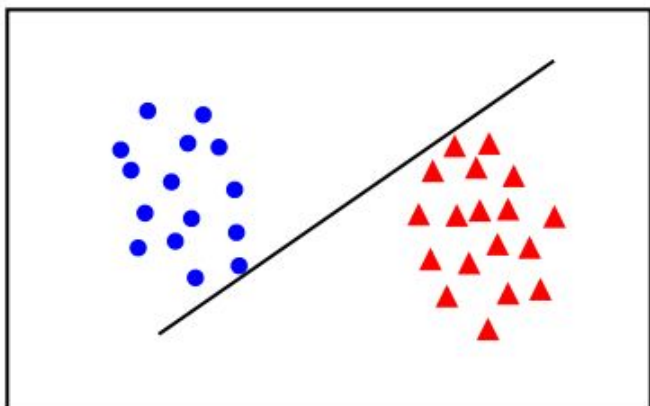
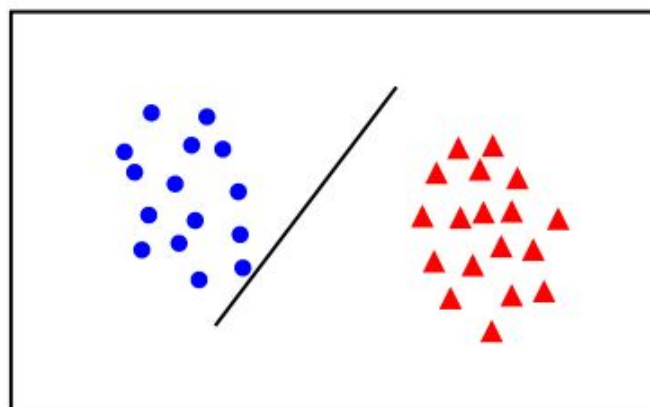
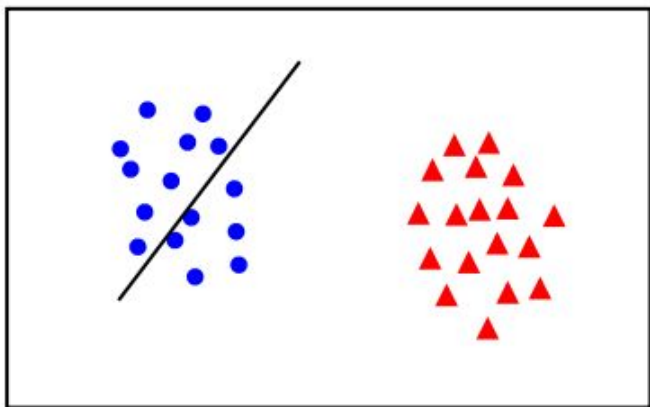
Lecture 9 | Take your own notes during lectures

Vinayak Abrol <abrol@iiitd.ac.in>



INDRAPRASTHA INSTITUTE *of*
INFORMATION TECHNOLOGY **DELHI**





Support Vector Machines (SVM)

- SVMs are supervised learning models with associated learning algorithms that analyze data for classification, regression and clustering analysis.
- Based on Vapnik–Chervonenkis (VC) theory a sub-branch of Statistical Learning theory.
 - What are (necessary and sufficient) conditions for consistency of a learning process based on the empirical risk minimization principle?
 - How fast is the rate of convergence of the learning process?
 - How can one control the rate of convergence (the generalization ability) of the learning process?
 - How can one construct algorithms that can control the generalization ability?

Support Vector Machines (SVM)

- Binary Classification
 - Non-Probabilistic
 - Probabilistic (Platt scaling)
 - Linear
 - Non-Linear (Kernel Trick)
- Regression
- Clustering



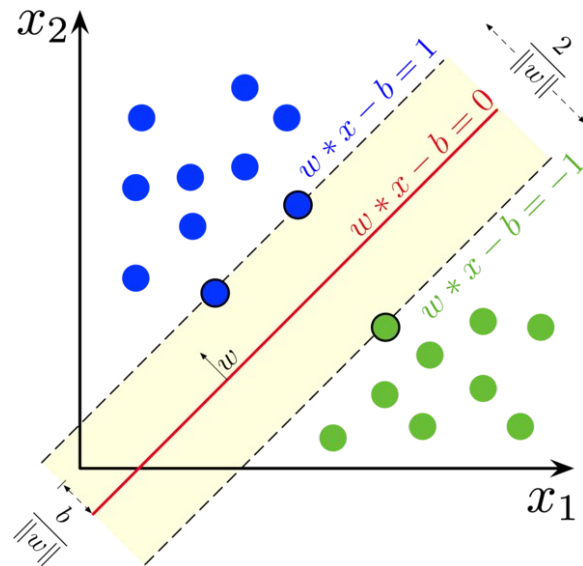
Coding
the SVM
algorithm in numpy



from sklearn
import svm

Support Vector Machines (SVM)

- SVM is all about maximum margin between classes!
 - What is this margin ?
 - How to ensure it exists?
 - How to optimize for it?

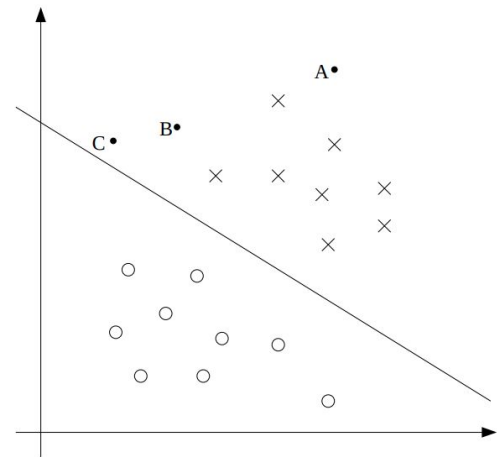
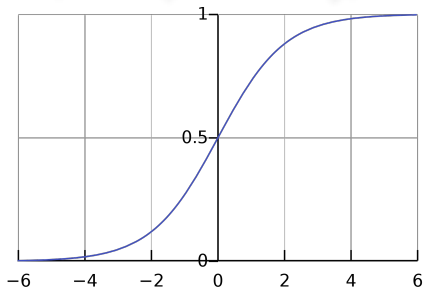


Support Vector Machines (SVM)

- SVM is all about maximum margin between classes!
 - What is this margin ?
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Consider logistic regression, where the probability $p(y = 1|x; \theta)$ is modeled by $h_{\theta}(x) = g(\theta^T x)$

We would then predict “1” on an input x if and only if $h_{\theta}(x) \geq 0.5$, or equivalently, if and only if $\theta^T x \geq 0$



$\theta^T x = 0$ is also called the separating hyperplane

SVM: Functional and Geometric margins

New Notations to be consistent with literature

Our classifier: $h_{w,b}(x) = g(w^T x + b)$ given input-output pairs (x_i, y_i) with $y \in \{-1, 1\}$

SVM: Functional and Geometric margins

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Our classifier will directly predict either 1 or -1, without first going through the intermediate step of estimating $p(y = 1|x; w, b)$ (which was what logistic regression did).

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- Functional Margin: $\hat{y} = y(w^T x + b)$
 - $y(w^T x + b) > 0$ means correct prediction
 - Large value means more confidence

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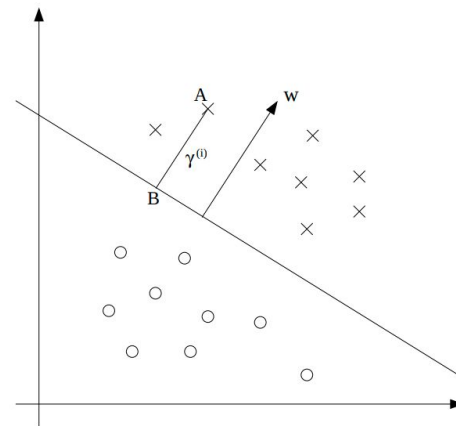
- Functional Margin: $\hat{y} = y(w^T x + b)$
 - $y(w^T x + b) > 0$ means correct prediction
 - Large value means more confidence
 - Scale Invariance: $h_{w,b}(x) = g(w^T x + b) = g(2w^T x + 2b)$
 - To avoid making functional margin arbitrarily large we use normalization
 - $(w / \|w\|_2, b / \|w\|_2)$
 - For the dataset (x_i, y_i) we are interested in the minimum $\hat{y}$

$$\hat{y}^{(i)} = y^{(i)}(w^T x^{(i)} + b)$$

SVM: Functional and Geometric margins

Consider Point A or $x(i)$

How can we find the value of $\gamma(i)$?

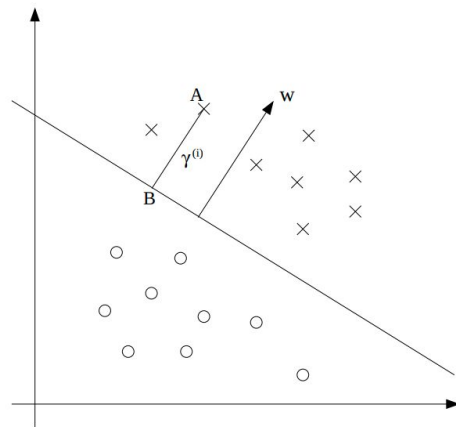


SVM: Functional and Geometric margins

Consider Point A or $x(i)$

How can we find the value of $\gamma(i)$?

- w is orthogonal (at 90°) to the separating hyperplane
- $w/\|w\|$ is a unit-length vector pointing in the same direction as w

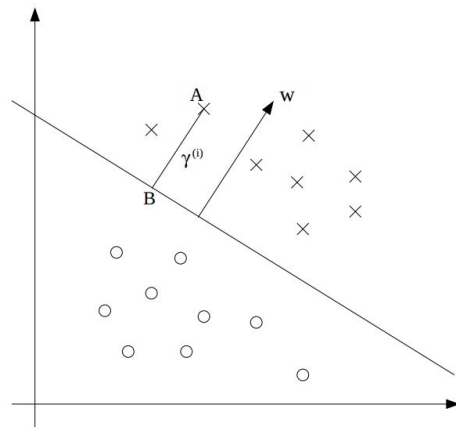


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- B is given by $x^{(i)} - \gamma(i) \cdot w/\|w\|$
- But B lies on our separating hyperplane $w^T x + b = 0$



SVM: Functional and Geometric margins

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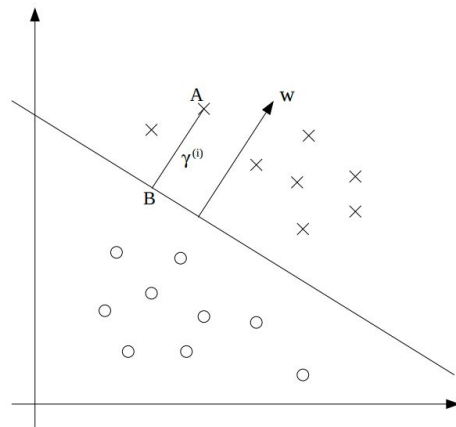
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$$w^T [x^{(i)} - \gamma^{(i)} w / \|w\|] + b = 0$$

$$w^T \left(x^{(i)} - \gamma^{(i)} \frac{w}{\|w\|} \right) + b = 0$$

$$\gamma^{(i)} = \frac{w^T x^{(i)} + b}{\|w\|} = \left(\frac{w}{\|w\|} \right)^T x^{(i)} + \frac{b}{\|w\|}$$



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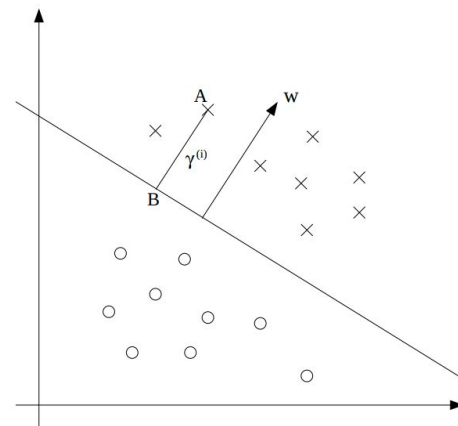
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Geometric Margin

$$\gamma = \min_{i=1,\dots,m} \gamma^{(i)}$$

$$\gamma^{(i)} = y^{(i)} \left(\left(\frac{w}{\|w\|} \right)^T x^{(i)} + \frac{b}{\|w\|} \right)$$



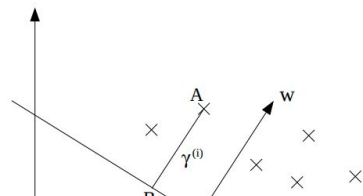
GM = FM
For $\|w\| = 1$

Scale Invariance
Exist!

SVM: Functional and Geometric margins

Consider Point A or $x^{(i)}$

How can we find the value of $\gamma(i)$?



Summary:

- w is orthogonal
 - $w/\|w\|$ is a unit vector
 - B is given by $x^T [x^{(i)} - \gamma^{(i)} w / \|w\|] + b = 0$
 - But B lies on the hyperplane
- The functional margin represents the lowest relative "confidence" of all classified points, while the geometric margin represents the smallest distance from to the hyperplane.

$$w^T \left(x^{(i)} - \gamma^{(i)} \frac{w}{\|w\|} \right) + b = 0$$

$$\gamma^{(i)} = \frac{w^T x^{(i)} + b}{\|w\|} = \left(\frac{w}{\|w\|} \right)^T x^{(i)} + \frac{b}{\|w\|}$$

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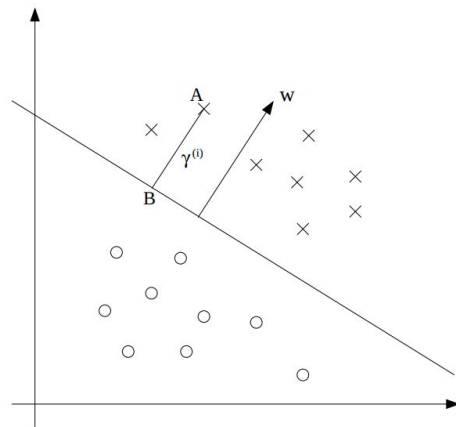
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Scale Invariance
Exist!

SVM: The Optimal Margin Classifier

Assumption: training set that is linearly separable.

$$\begin{array}{ll}\max_{\gamma, w, b} & \gamma \\ \text{s.t.} & y^{(i)}(w^T x^{(i)} + b) \geq \gamma, \quad i = 1, \dots, m \\ & \|w\| = 1.\end{array}$$

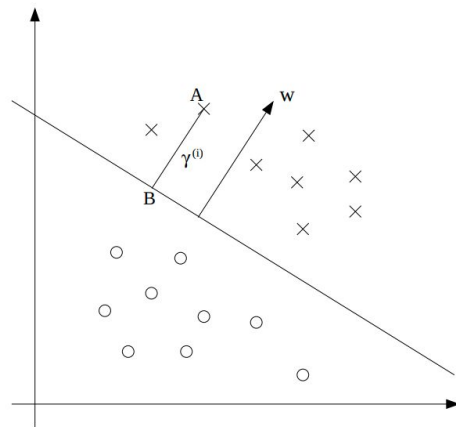


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Solve this and we are done!



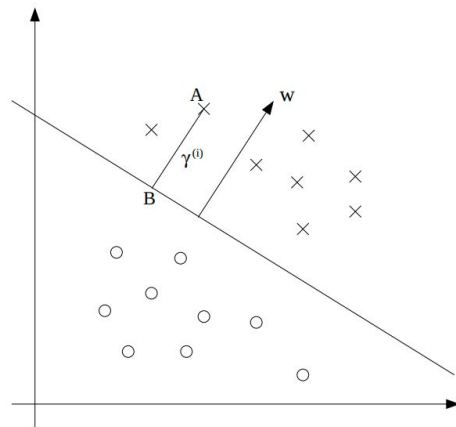
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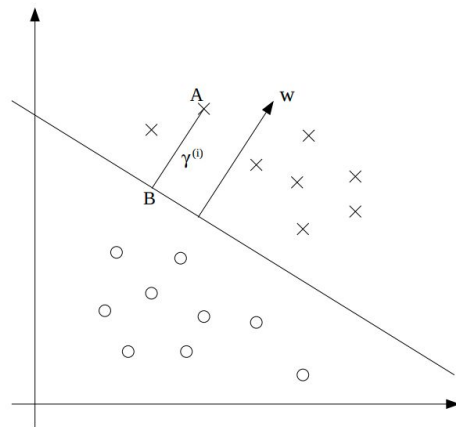
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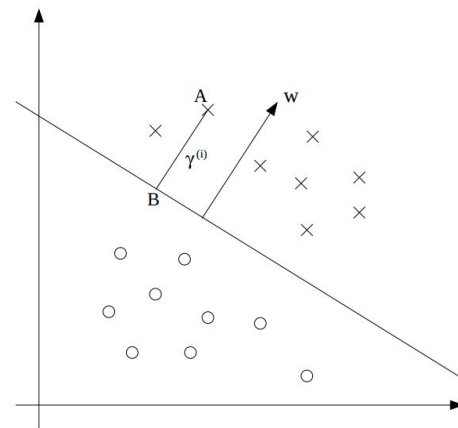
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$$\hat{\gamma} / \|w\| = 1 / \|w\|$$

$$\begin{aligned} \min_{\gamma, w, b} \quad & \frac{1}{2} \|w\|^2 \\ \text{s.t.} \quad & y^{(i)}(w^T x^{(i)} + b) \geq 1, \quad i = 1, \dots, m \end{aligned}$$

Recall the scaling property of functional margin!

SVM: The Optimal Margin Classifier

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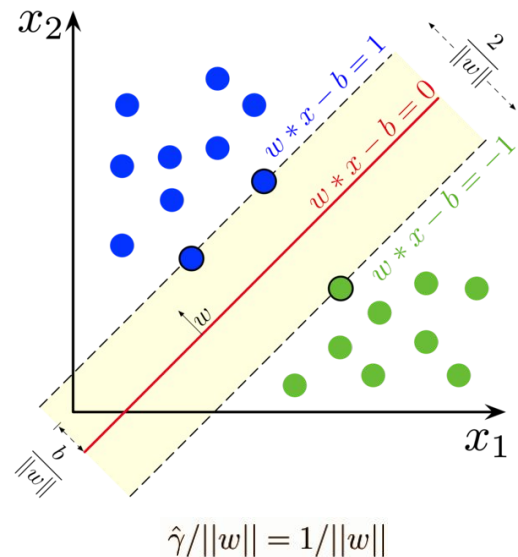
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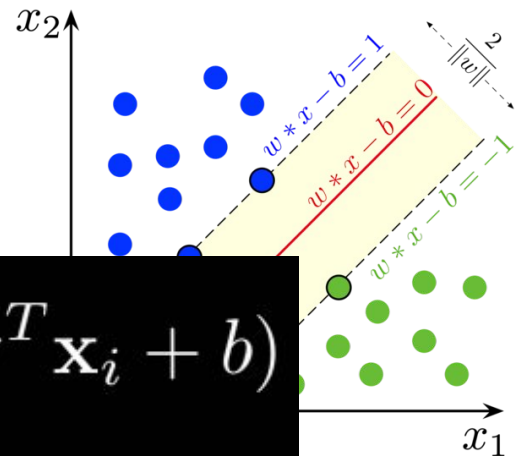
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Solve

But

$$\operatorname{argmax}_{(w, b)} \frac{1}{\|w\|} \quad \text{s.t.} \quad \min_{i \in [1:n]} y_i(w^T x_i + b)$$



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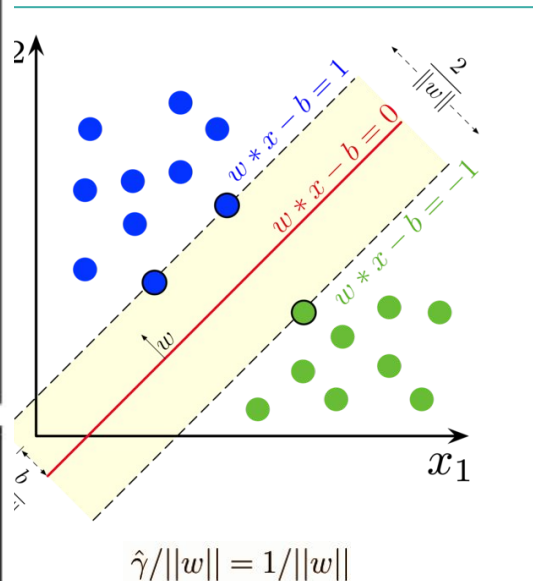
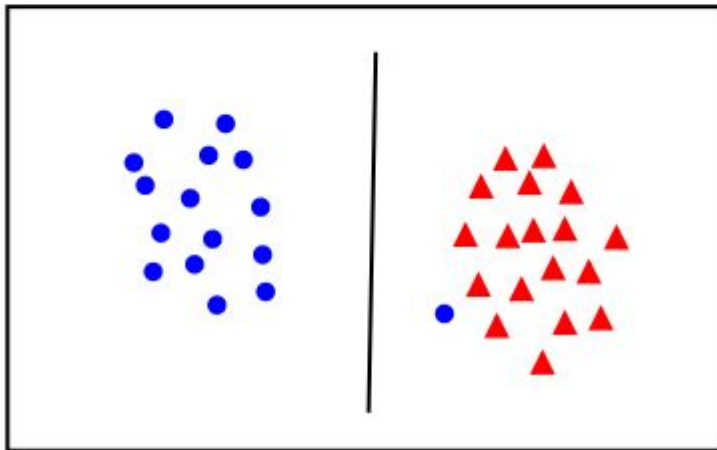
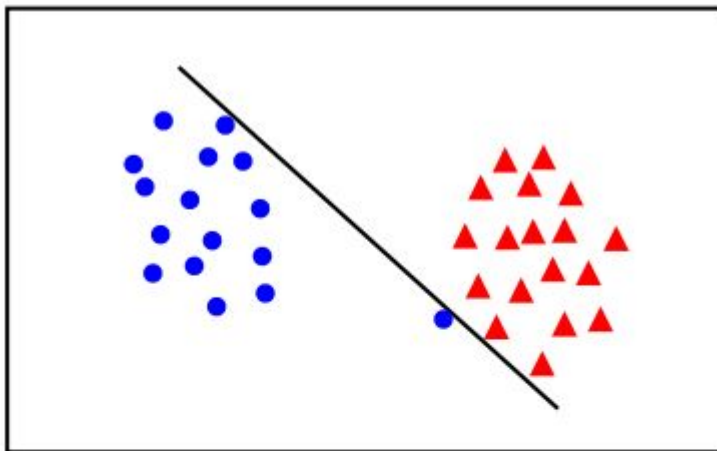
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SVM: Maths Looks Heavy?

To be continued.....



Layman's approach is not always a good one, especially for students who want to master the underlying logic of an algorithm.

Then you have to, I hate but I gotta say, dive deep into the hard math and understand the formulas.

You can't learn unless you try it yourself.